

Consider the linear time-invariant system

$$x(k+1) = \underbrace{\begin{bmatrix} 1/2 & -1/4 \\ 1/4 & 1/2 \end{bmatrix}}_A x(k) + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_B u(k)$$

subject to state constraints $\|x(k)\|_\infty \leq 1$ and input constraints $|u(k)| \leq 1$. The MPC formulation for the general reference tracking case with $Q = I_2$ and $R = 1$ is given as follows.

$$\begin{aligned} \underset{U}{\text{minimize}} \quad & (x_N - x_s)^\top P (x_N - x_s) + \sum_{i=0}^{N-1} (x_i - x_s)^\top (x_i - x_s) + (u_i - u_s)^\top (u_i - u_s) \\ \text{subject to} \quad & x_{i+1} = Ax_i + Bu_i, \\ & x_i \in \mathcal{X} = \{x \mid \|x\|_\infty \leq 1\}, \\ & u_i \in \mathcal{U} = \{u \mid |u| \leq 1\}, \\ & x_N \in \{x_s\} \oplus \mathcal{X}_f, \\ & x_0 = x(k). \end{aligned} \tag{1}$$

Exercise 1 (Regulation Case)

Consider first the regulation case, i.e., $x_s = 0$ and $u_s = 0$, and a zero terminal control law $\kappa_f(x) = 0$. Show that the choice of

- terminal set $\mathcal{X}_f = \{x \mid x^\top x \leq 1\}$, and
- terminal cost weighting $P = aI$, with scalar parameter $a > 0$, that solves the Lyapunov equation $A^\top P A - P = -I$,

results in a recursively feasible and asymptotically stabilizing MPC controller.

Exercise 2 (Reference Tracking)

Consider the reference tracking problem with the system output $y(k) = Hx(k) \rightarrow r$ with $H = [1 \ 0]$ and reference $r = 0.25$.

- Compute the resulting steady state x_s and corresponding input u_s .
- Modify the terminal set $\mathcal{X}_f$ such that the MPC problem (1) is recursively feasible and accurate tracking is guaranteed.

Exercise 3 (Disturbance Estimation)

Consider now that the system is subject to time-invariant disturbances d

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) + d \\ y(k) &= x(k) \end{aligned}$$

where we assume to be able to measure the full state $x(k)$.

- Derive the estimation-error dynamics for the augmented state $\tilde{x} = [x(k) \ d]^\top$ under a linear estimator L
- Design a stable estimator gain L in Matlab (e.g., using pole placement)
- Simulate the autonomous system from $x(0) = \hat{x}(0) = [1 \ 1]^\top$ with $d = [0.1 \ -0.2]^\top$ and $\hat{d}(0) = [0 \ 0]^\top$.